

BUILDING QUANTUM TELEPORTATION

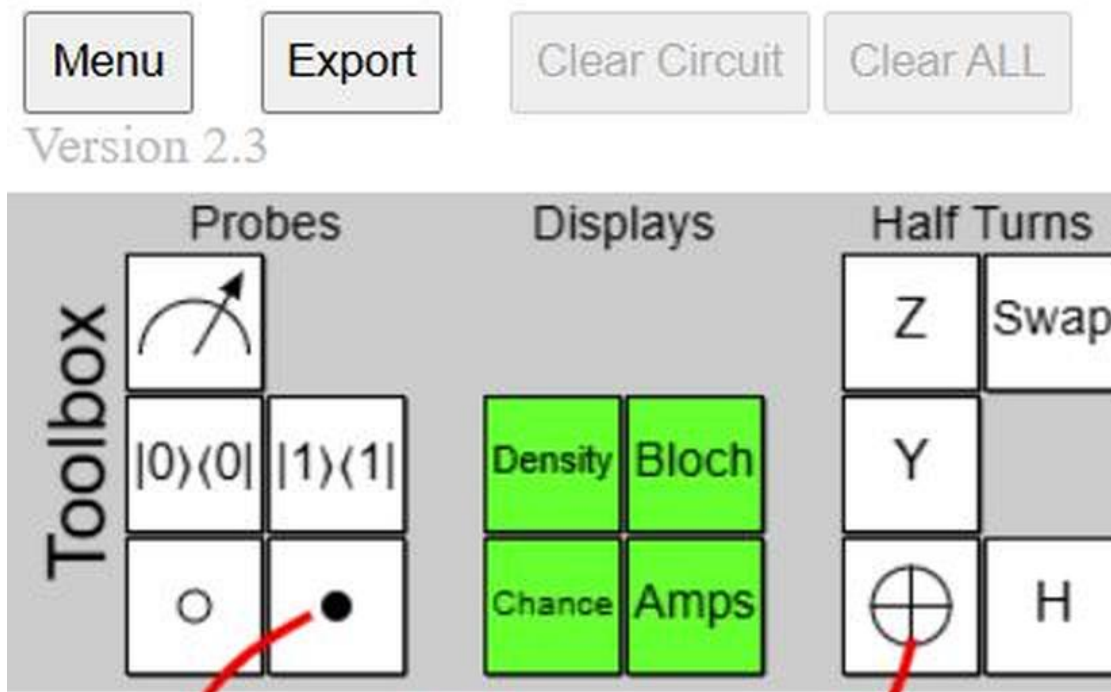
A step-by-step Quirk build — Station B

You're going to build a real quantum circuit, from a totally blank canvas, that teleports the state of one qubit onto another — using nothing but entanglement and a phone call. Quirk is a free drag-and-drop simulator, so there's nothing to install: open the link, and every gate you need lives in the toolbox at the top of the screen.

Go to: quirk-e.dev (use this link: it has the message gate pre-loaded)

Before you start: the toolbox

Every gate you'll use today lives in the same corner of the toolbox. Take a second to spot all five before you begin building.



The toolbox, top-left corner. This is everything you need for the whole build.

Gate	Where to find it	What it does
H	"Half Turns" column, bottom-right box	Puts a qubit into superposition — 50/50 between 0 and 1
$\oplus$ (circle-plus)	"Half Turns" column, bottom-left box	The receiving end of a CNOT — flips its qubit only if the control is 1
• (control dot)	"Probes" column, bottom-right box	The sending end of a CNOT — stack it directly above a $\oplus$ on another wire
Z	"Half Turns" column, top-left box	Flips the sign of a qubit's 1-state — invisible until it's controlled
Measure	"Probes" column, top-left box (the meter icon)	Collapses a qubit to a definite 0 or 1 and reads it out

Adding wires

A blank Quirk circuit starts with 2 wires. You'll need 3 for this build — one for the message, and one each for the two entangled halves. To add a wire, just drag any gate to the empty row directly below the bottom wire — Quirk adds the wire for you automatically the moment you drop it there. No button, no menu.

The build — in three stages

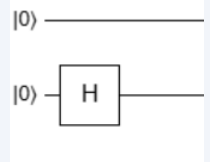
Stage 1 — Make an entangled pair (~5 min)

1

Put wire 2 into superposition

Drag an H gate onto wire 2 (the middle wire), first column.

This is Alice's half of what will become the entangled pair.

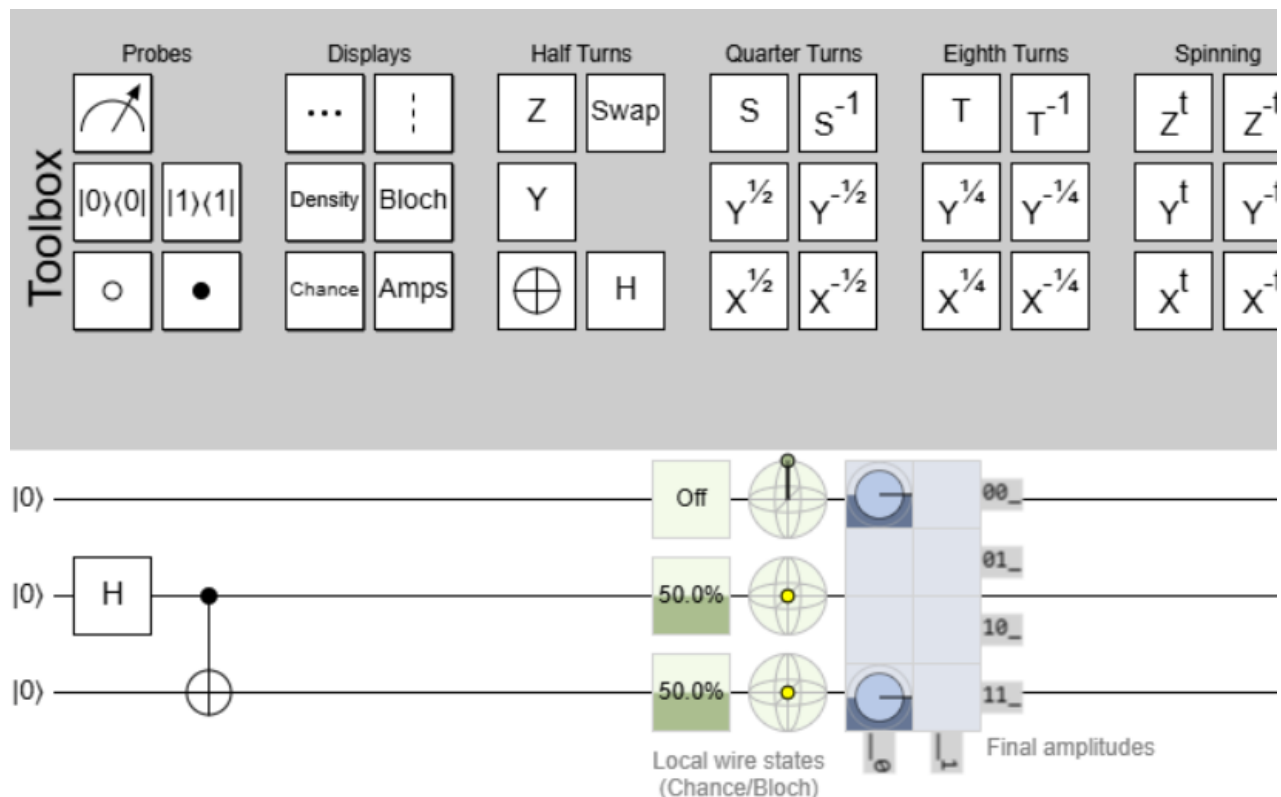
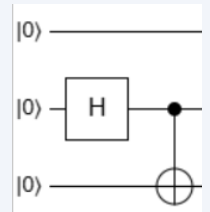


2

Entangle wires 2 and 3

Drag a control (•) onto wire 2, one column to the right of the H. Then drag a $\oplus$ onto wire 3, in that same column, directly below the •.

Wires 2 and 3 are now entangled — measuring one instantly tells you about the other, however far apart they are.



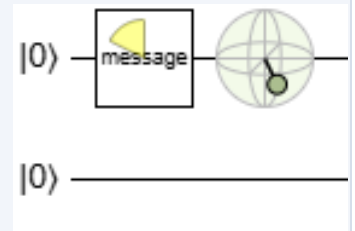
Stage 1 complete: an entangled pair on wires 2 and 3.

Stage 1 — Prepare a message and read Alice's half (~5 min)**1****Check that you have a message block and visualise it**

The message gate has been given for you at the start- it should have a little circle that spins around.

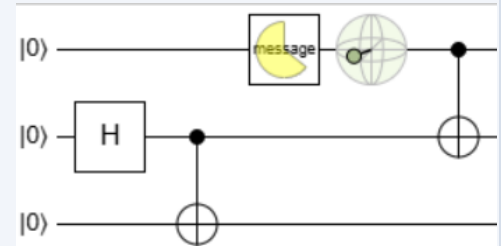
Add a Bloch display after it to visualise the value

This is the “secret” quantum state you’re about to teleport - for this build it’s just superposition (value between 1 and 0), but it could be any qubit state.

**2****Link the message to Alice's half**

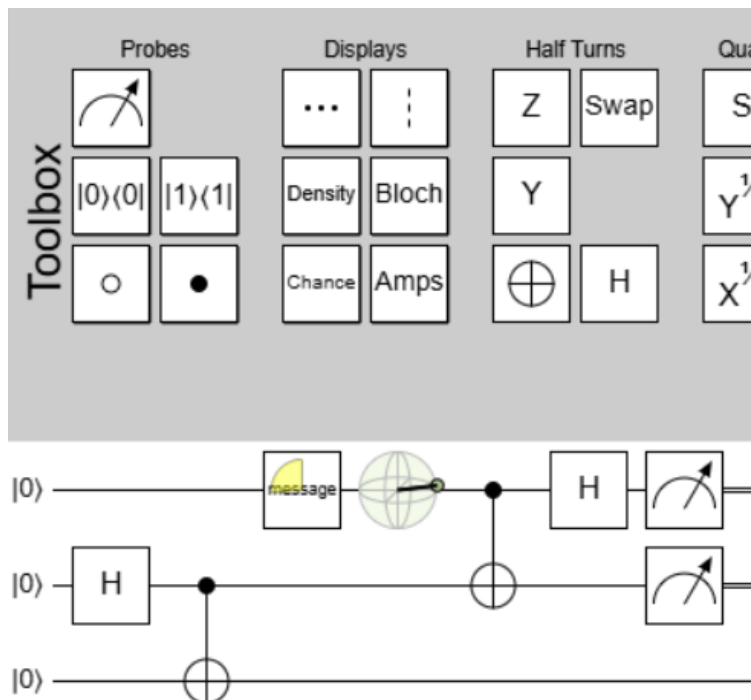
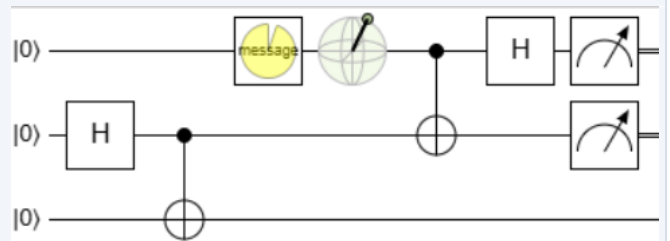
Drag a control (●) onto wire 1 and a $\oplus$ onto wire 2, in the same column, just after Stage 1's gates.

This mixes the message into Alice's half of the entangled pair.

**3****Finish Alice's measurement**

Drag an H onto wire 1, then drag two Measure gates (the meter icon) — one onto wire 1, one onto wire 2 — in the same column, right after.

Alice now has two classical bits — 0s and 1s, nothing spooky about them. These are what she'd have to phone through to Bob.



Stage 2 complete: wire 1 (message) and wire 2 (Alice's half) both measured, collapsing to ordinary bits.

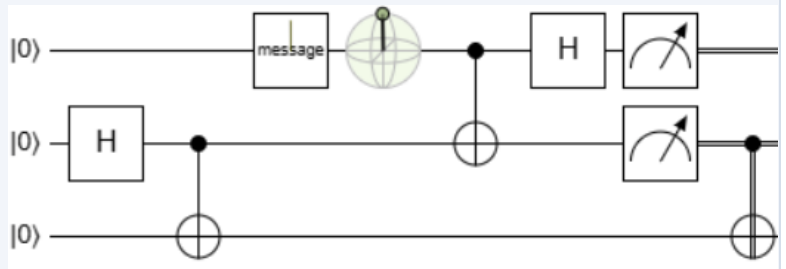
Stage 3 — Bob's corrections (~5 min)

7

Apply the flip, if needed

Drag a control (●) onto wire 2 (right after the Measure gates) and a $\oplus$ onto wire 3, in that same column.

This says: "if Alice's second bit was 1, flip Bob's qubit."

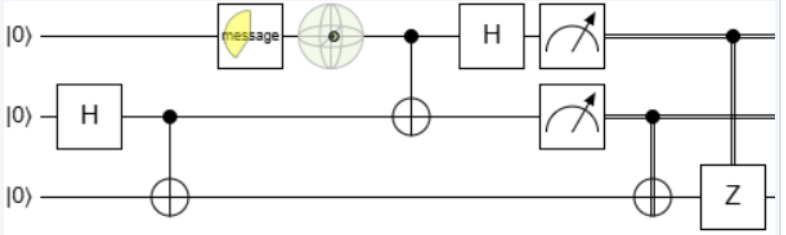


8

Apply the phase, if needed

Drag a control (●) onto wire 1 and a Z onto wire 3, one column further right.

This says: "if Alice's first bit was 1, apply a Z to Bob's qubit."

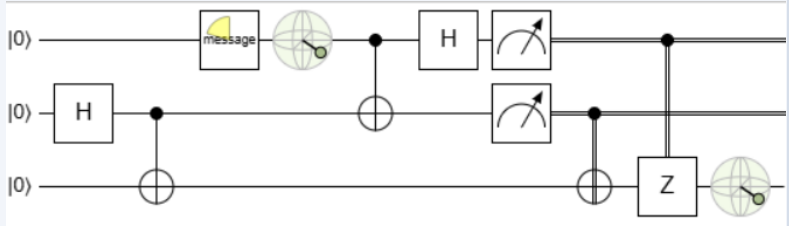


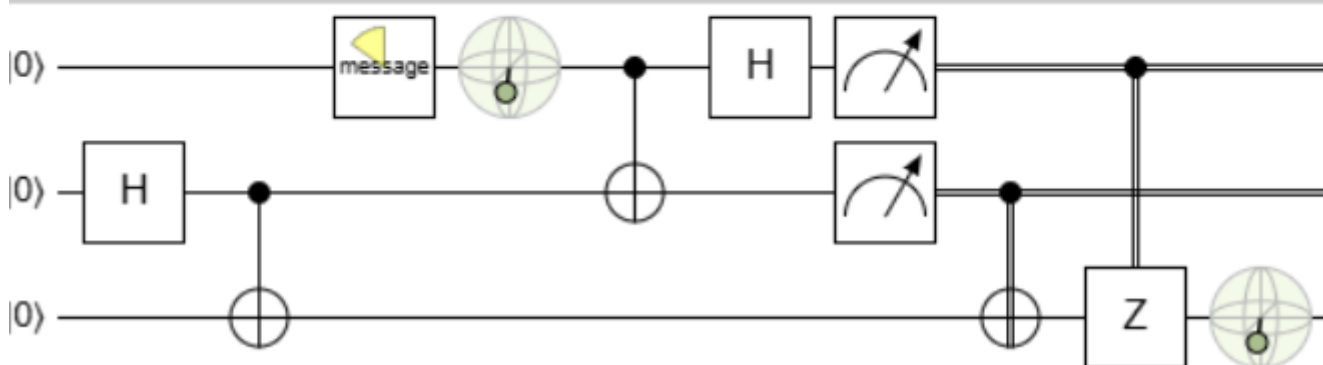
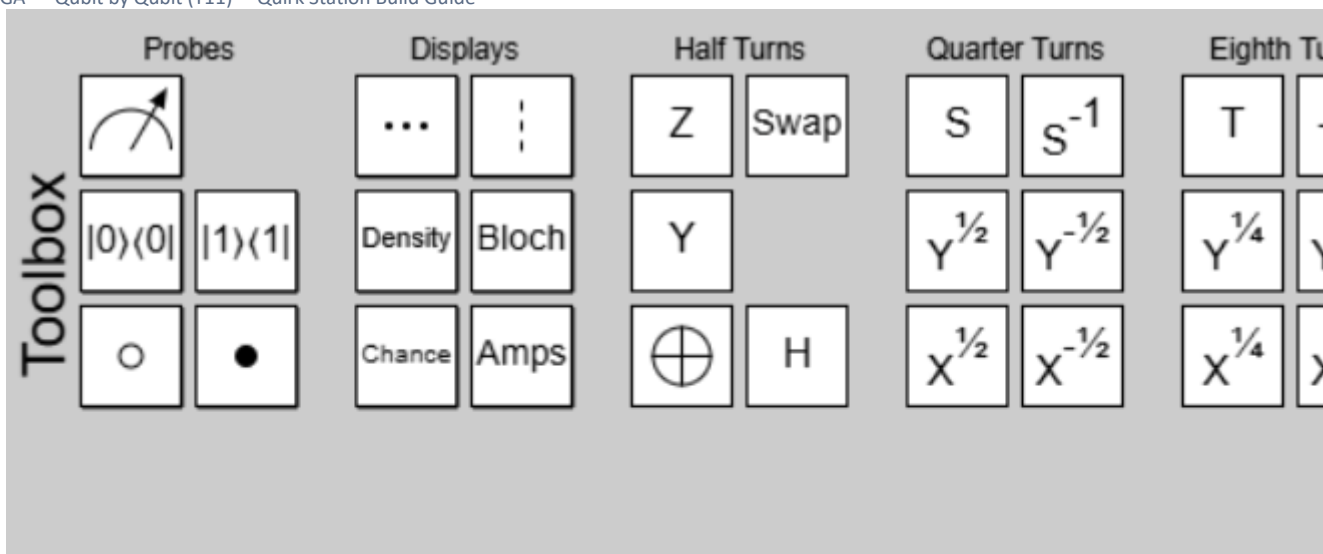
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See the result

Drop a final Bloch sphere on wire 3, after everything else.

Compare it to the Bloch sphere you'll add on wire 1 back at the start (Stage 2, step 4) — once Bob applies his corrections, his sphere matches the original message exactly.





The finished circuit: message prepared on wire 1, entangled pair on wires 2–3, Alice's Bell measurement, Bob's two classically-controlled corrections, and a final Bloch sphere on wire 3 matching the original message.

Talk it through

- Cover the classical wires (2 and 3, after the measurement) with your hand - could Bob get the right answer without them? Try deleting the two controlled gates and see what his Bloch sphere shows instead.
- Nothing here sends information faster than light — the two measured bits still have to travel to Bob by an ordinary, speed-of-light channel. That's a phone call, a text, an email – classic, current communication techniques.
- The original qubit on wire 1 is destroyed by measurement in Stage 2 — this isn't copying, it's teleporting. Ask: why can't you have two copies of the same qubit at once?

Challenge Tasks

- **Break it on purpose** — delete just one of Bob's two corrections and see which part of the state goes wrong.
- **No entanglement, no teleportation** — remove the entangling gates and watch it fail, which makes the "why does this need entanglement" point concrete.
- **Chain two teleports** — a genuine stretch task for fast finishers: relay the state across two hops using two extra wires, which mirrors how real quantum repeaters work.